

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

INTRO TO CRITICAL INFRASTRUCTURE FOR IT LEARNERS

LAB 05 — BUILDING A SCADA HUMAN MACHINE INTERFACE

Student Edition

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Lab 05: Building a SCADA Human Machine Interface

Course: Intro to Critical Infrastructure for IT Learners | **Estimated Duration:** 4–5 hours

Submission deadline: As published in ICDFA E-Campus.

Authorization & Safety: Perform all activities only on the ICDFA-provided virtual machines, packet captures and isolated training networks. Do not bridge these VMs to production networks, scan external systems, or reuse credentials observed in training data.

Learning Outcomes

- Apply basic high-performance HMI and situational-awareness principles.
- Create a tank, level indication, valve and process piping in FUXA.
- Connect the HMI to the OpenPLC tank-fill simulation.
- Bind HMI objects to PLC/Modbus tags and configure simple animation/state indication.
- Demonstrate that the HMI reflects the running process correctly.

Laboratory Environment

System	Expected Address	Account	Password
PLC VM	172.16.100.137	plc	password
SCADA VM	172.16.100.141	scada	password

Note: If the instructor-provided VM image uses updated addressing or credentials, use the values published in E-Campus. Do not alter addressing unless instructed.

Pre-Lab Checklist

- Read the complete lab before beginning.
- Confirm you can access the required VM(s) and that virtualization support is enabled.
- Use the instructor-specified isolated/host-only network.
- Create or restore the clean starting snapshot for this lab.
- Prepare a report document with your name, registration number, course, lab number and date.
- Capture evidence as you progress rather than repeating the lab at the end.

Procedure

Step 1. Start the PLC and SCADA VMs in the isolated laboratory network. Start FUXA on the SCADA VM and open <http://127.0.0.1:1881/editor>.

Step 2. Create a New Project named with your registration number and LAB05, for example 2026XXXX_LAB05.

Step 3. Set a neutral gray background and enable the grid while arranging process graphics.

Step 4. From the Process Engineering objects, place a tank graphic. Add a bar gauge over/inside the tank and set its engineering range to 0–10 ft.

Step 5. Build a simple motor-actuated fill valve using the available valve/shape objects. Add a pipe from the valve to the tank. Keep the normal-state design visually simple.

Step 6. On the PLC VM, load/start the provided TankFillSim.st program in OpenPLC Runtime and confirm the status is Running.

Step 7. In FUXA, create or edit the Modbus TCP connection to the PLC at 192.168.1.10:502.

Step 8. Create/verify the Tank Level tag mapped to the OpenPLC input-register location used by the simulation (%IW80 / Input Register 80 in the teaching image).

Step 9. Create/verify the Tank Valve Open tag mapped to the OpenPLC discrete output used by the simulation (%QX0.0 / Coil 0 in the teaching image).

Step 10. Bind the bar gauge to the Tank Level tag. Configure Min = 0 and Max = 10.

Step 11. Bind the valve-state graphic to Tank Valve Open. Configure the normal/closed representation for value 0 and the open representation for value 1.

Step 12. Use a restrained high-performance style: neutral normal states, clear labels, and attention-drawing color only where it communicates a meaningful abnormal/alarm condition.

Step 13. Enable the PLC/FUXA connection and run the HMI. Observe the tank filling/drainage and confirm the valve-state graphic changes with the process.

Step 14. Save/export the FUXA project and capture evidence of both the design view and running HMI.

Required Evidence & Submission

- Screenshot of the HMI in design/edit mode.
- Screenshot of the running HMI with a visible tank-level value.
- Screenshot of the configured PLC connection/tags with sensitive passwords excluded.
- The exported FUXA project file.
- A short design rationale explaining how the interface supports situational awareness.

Design Rationale

The HMI was designed using high-performance HMI principles to improve situational awareness. A neutral gray background reduces visual clutter and operator fatigue. The tank graphic and level indicator provide a clear representation of the process state, while the valve graphic shows the current valve position. Colors are used sparingly so that abnormal conditions can be immediately recognized. The layout follows the process flow from valve to tank, making it easy for operators to understand system status and identify changes in tank level or valve operation.

Knowledge Check

1. What is a P&ID and how does it influence HMI layout?

Response: A P&ID (Piping and Instrumentation Diagram) is an engineering drawing that shows process equipment, piping, instruments, control devices, and signal flows. It influences HMI layout by providing a structured representation of the process, allowing operators to see equipment and process flow in a familiar and logical arrangement.

2. Why do high-performance HMIs avoid unnecessary animation and excessive color?

Response: High-performance HMIs avoid unnecessary animation and excessive color because they can distract operators and hide important information. Using neutral colors for normal conditions and highlighting only abnormal situations helps operators quickly identify problems and make effective decisions.

3. How does the Tank Level tag differ from the Tank Valve Open tag in data type and behavior?

Response: The Tank Level tag represents a continuously changing numerical value and is typically stored as a REAL or analog value. The Tank Valve Open tag represents a discrete state and is stored as a BOOL value, indicating whether the valve is open (1) or closed (0).

4. What could happen operationally if an HMI displays stale or incorrect PLC data?

Response: If an HMI displays stale or incorrect PLC data, operators may make incorrect decisions based on inaccurate process information. This can result in process interruptions, equipment damage, safety hazards, false alarms, or failure to respond to actual abnormal conditions.